AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Mitosis, meiosis and reproduction

Cells divide for two reasons: to make identical copies for growth and repair, and to make gametes for sexual reproduction. This page covers linkage, meiosis and the variation it creates, how gametes are built and fuse in mammals and flowering plants, the cell cycle and mitosis, the root tip squash and the mitotic index.

SPEC 3.9 Gene loci and linkage

WHAT THIS SPEC POINT ASKS

(i) Know that a locus is the position of a gene on a chromosome. (ii) Understand linkage of genes on the same chromosome.

Key words

- **Gene**: a section of DNA that codes for a polypeptide.
- **Allele**: an **alternative form of a gene** (a different base sequence at the same locus).
- **Locus**: the **position of a gene** on a chromosome. Each gene has a fixed locus.
- **Homologous chromosomes**: a pair (one from each parent) with the **same genes at the same loci**, though they may carry different alleles.

Linkage

- Genes whose loci are on the **same chromosome** are **linked**: they tend to be **inherited together**, because they move into the gametes as one unit in meiosis. They don't assort independently.
- Linked alleles can be separated only by **crossing over** between **non-sister chromatids** of homologous chromosomes in **prophase I** (3.10). Gametes with the new combinations are **recombinants**.
- Crossing over is rare at any one point, so recombinants are always **fewer** than the parental types.
- The **closer** two loci are, the **less likely** a crossover falls between them, so the more tightly they are linked.
- Sex linkage (Unit 1) is a special case of linkage: genes on the X chromosome are inherited together with sex.

WORKED EXAMPLE

gametes from linked genes

A parent is AaBb. One chromosome carries **A** and **B**; its homologue carries **a** and **b**.

If the genes were on **different** chromosomes: four gamete types, **25% each**.

Linked, with crossing over between them in 10% of gametes:

- **AB 45%** and **ab 45%** (parental types: linkage kept)
- **Ab 5%** and **aB 5%** (recombinants: made by crossing over)

The more the results depart from 25% each, the closer the loci.

MARK-SCHEME WORDING

- Locus: "{location / position} of a {gene / allele} on a chromosome" (chromatid accepted).
- Allele: "an alternative form of a gene".
- Linkage: "the genes are on the **same chromosome**" **and** "have a higher chance of being **inherited together** / not separated by crossing over". Stopping at "same chromosome" loses the second mark.
- Closer loci: "alleles are closer together, **therefore** crossing over between them is less likely".
- Gamete data (not 25% each): "(due to) linkage"; "A and B on the same chromosome"; "AB and ab inherited together"; "Ab and aB formed by **crossing over**" (recombinants).

Often asked as: define locus or allele; explain why two alleles are inherited together more often than others; explain gamete percentages that aren't 25%; decide which genes are linked from chromosome data.

COMMON MISTAKE

Writing about dominance, mutation or independent assortment when linkage is asked.

Crossing over between **different** chromosomes (e.g. chromosome 1 and chromosome 4). It happens only between **homologous** chromosomes.

Confusing locus with chiasma, centromere or centrosome. Using "gene" when you mean "allele".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Working out which recombinant combinations a single crossover can make across **three loci** (WBI12 Jun 2021).
- **Linked and unlinked genes together**: two genes on chromosome 1 are linked and inherited together; a third on chromosome 4 assorts independently of both (WBI12 Jun 2022).

SPEC 3.10 Meiosis and genetic variation

WHAT THIS SPEC POINT ASKS

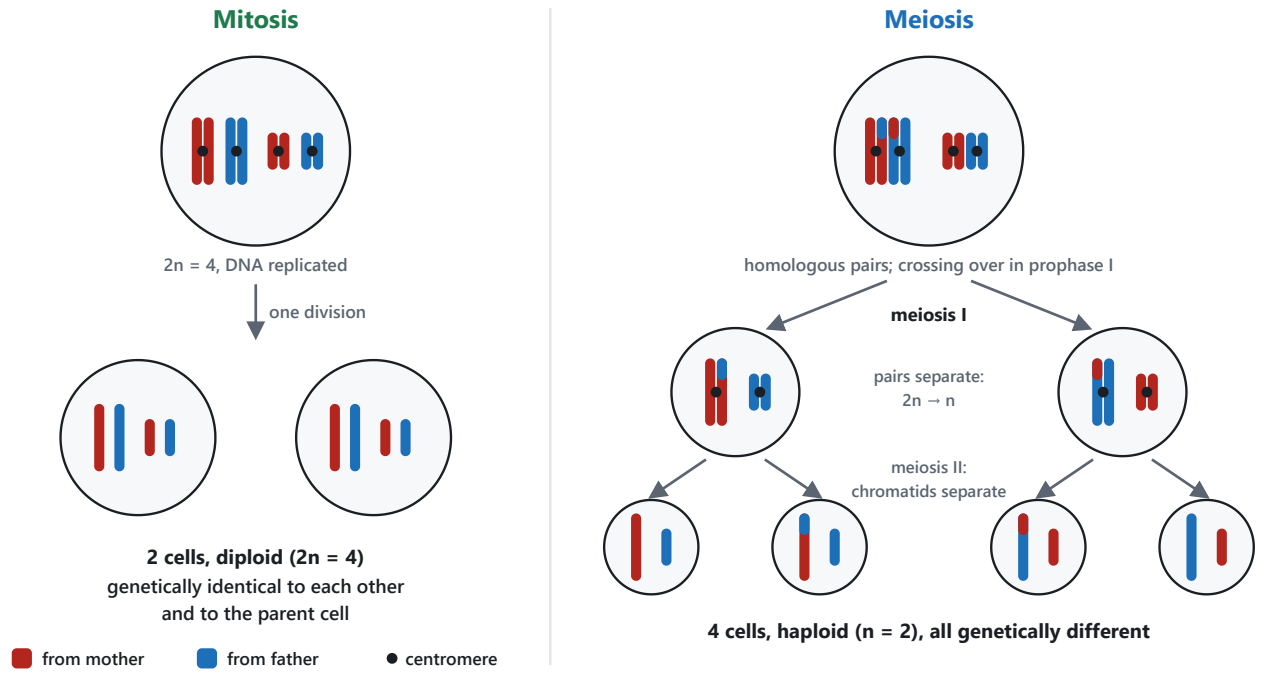
Understand how meiosis makes genetically different gametes, through independent assortment of chromosomes at metaphase I and crossing over between chromatids in prophase I. (You don't need the names of the stages of prophase.)

Why meiosis

- Meiosis is a **reduction division**: one **diploid** (2n) cell divides **twice** to make **four haploid** (n) cells, each **genetically different**.
- Halving the number means fertilisation (fusing two haploid nuclei) **restores** the diploid number every generation.
- Before meiosis, DNA is replicated (S phase), so each chromosome is two identical **sister chromatids** joined at a **centromere**.

The two divisions

Stage	Meiosis I: **homologous pairs** separate	Meiosis II: **sister chromatids** separate
Prophase	chromosomes condense; homologous chromosomes **pair up** (bivalents); **crossing over** at chiasmata; nuclear envelope breaks down; spindle forms	chromosomes condense again; new spindles form
Metaphase	**pairs** line up on the equator, each pair's orientation random (**independent assortment**)	single chromosomes line up on the equator
Anaphase	**homologous chromosomes** pulled to opposite poles; chromatids stay joined	**centromeres divide**; sister chromatids pulled apart
Telophase	two **haploid** nuclei (each chromosome still two chromatids)	**four haploid** nuclei



A cell with 2n = 4. Mitosis: one division, two identical diploid cells. Meiosis: crossing over (swapped chromatid ends) and independent assortment (which long and short chromosome go together), then two divisions give four haploid cells that all differ.

Source 1: crossing over (prophase I)

1. Homologous chromosomes pair up closely.
2. **Non-sister chromatids** (one maternal, one paternal) wrap round each other, break at the same points and **rejoin** to the other chromatid. The crossing points are **chiasmata**.
3. Sections, and the **alleles** on them, are **exchanged**. The chromatids now carry **new combinations of alleles** (recombinant chromatids).

Source 2: independent assortment (metaphase I)

1. Homologous pairs line up on the equator. Whether the maternal or paternal chromosome of each pair faces a given pole is **random**, and each pair lines up independently of the others.
2. So each gamete gets a **different mixture of maternal and paternal chromosomes**, and so a different combination of alleles.
3. Number of combinations = $\mathbf{2^n}$, where n is the **haploid** number.

WORKED EXAMPLE

2^n

A fruit fly has 2n = 8, so n = 4: 2^4 = **16** combinations of chromosomes in its gametes.

Humans: n = 23, so 2^{23} = **8 388 608** combinations, before crossing over is even counted.

Random fertilisation: any sperm can fuse with any egg, so one couple could make $2^{23} \times 2^{23} \approx 7 \times 10^{13}$ different chromosome combinations in a child.

Meiosis v mitosis

	Mitosis	Meiosis
Number of divisions	one	**two**
Cells made	**two**	**four**
Chromosome number	stays the same (2n → 2n)	**halved** (2n → n)
Genetic make-up	daughters **identical** to each other and to the parent	daughters **all different**
Homologous chromosomes pair?	no	**yes**, in prophase I
Crossing over	no	**yes**, prophase I
Metaphase	chromosomes line up singly	**pairs** line up (metaphase I); singles (metaphase II)
Where	body cells: growth, repair, asexual reproduction	making **gametes** (testes, ovaries; anthers, ovules)

EXAM TIP

Read the question's wording. "Offspring" means add **random fertilisation**. "Separation of chromosomes" means **independent assortment only**: crossing over earns nothing there. Always say "prophase **I**" and "metaphase **I**".

MARK-SCHEME WORDING

- "Crossing over results in different **allele** combinations"; "independent assortment results in different combinations of chromosomes / alleles in the gametes". "Gene" for allele is rejected.
- Crossing over: "sections of chromatids **exchanged** at **chiasmata**" "between **non-sister** (maternal and paternal) chromatids". Sister chromatids is rejected.
- Offspring variation: independent assortment, "a mixture of maternal and paternal chromosomes / different combination of alleles", and "**random fertilisation**" (which gametes fuse is random).
- Egg v body cell: "23 chromosomes / haploid"; "so fertilisation forms a **diploid** zygote"; different "due to crossing over / independent assortment".
- DNA graph for meiosis: same shape as mitosis to start, then "a **further division**" so DNA falls in **two steps** (4 → 2 → 1 a.u.).

- Metaphase of mitosis v meiosis: similarities "chromosomes line up on the equator", "centromeres attach to spindle fibres"; differences "two metaphases in meiosis", "**homologous pairs** line up in metaphase I but single chromosomes in mitosis", "independent assortment in metaphase I". Full marks need a similarity.

Often asked as: explain how meiosis produces variation; complete a diagram after crossing over; draw the gametes from a given cell; sketch or explain a DNA-content graph; compare metaphase in mitosis and meiosis.

COMMON MISTAKE

Writing "fusion of egg and sperm **cells**". It is the **nuclei** that fuse.

Drawing a mitosis graph for meiosis, or dropping straight from 4 to 1 a.u.

Forgetting random fertilisation when the question says "offspring".

Ignoring the drawing or chromosome number in the question (e.g. "26 chromosomes": half come from each parent).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Non-disjunction**: a pair of chromosomes (meiosis I) or chromatids (meiosis II) fails to separate, so one gamete gets an **extra** chromosome and another gets **none**. In Patau syndrome the zygote has three copies of chromosome 13 (WBI12 Oct 2024).
- **Hybrid sterility**: chromosomes from two different species can't **pair up** in prophase I, so meiosis fails and no gametes form. "An odd number of chromosomes" isn't credited (WBI12 Jun 2022).
- An error in crossing over **inside a gene** can create a new allele, or duplicate a gene (WBI12 Jan 2023, Oct 2025).

SPEC 3.11 Mammalian gametes

WHAT THIS SPEC POINT ASKS

Understand how sperm and egg cells are specialised for their functions, including the acrosome in sperm and the zona pellucida in the egg.

Sperm: built to swim and to get in

Feature	How it helps
Haploid nucleus (in the head)	fuses with the haploid egg nucleus to make a **diploid** zygote
Acrosome: a vesicle of **hydrolytic enzymes** (e.g. **acrosin**) over the front of the head, made by the Golgi	the enzymes **digest a path through the zona pellucida**
Receptors / proteins on the head membrane	**bind** to the zona pellucida
Midpiece packed with **mitochondria**	**aerobic respiration** releases **ATP** for the flagellum
Flagellum (tail) of microtubules	beats to make the sperm **swim** to the egg
Small, streamlined, very little cytoplasm	less **resistance**, so it swims fast; huge numbers can be made

Egg (secondary oocyte): built to feed the embryo and let in only one sperm

Feature	How it helps
Haploid nucleus	fuses with a sperm nucleus to restore the diploid number
Very large, with lots of **cytoplasm**: **lipid droplets**, protein, mitochondria, ribosomes	nutrients and ATP for the early embryo until it implants
Zona pellucida: a thick layer of **glycoprotein** just **outside** the cell surface membrane	sperm **bind** to receptors in it (which sets off the acrosome reaction); after fertilisation it **hardens** to keep other sperm out
Cortical granules: vesicles of enzymes just under the membrane	released in the **cortical reaction** (3.12) to harden the zona
Follicle cells (outside the zona)	protect the egg

- A human egg is about 0.1 mm across, visible to the naked eye; a sperm head is only about 5 μm long.
- Many sperm, few eggs: most sperm never reach the egg, and they **compete** to fertilise it.

MARK-SCHEME WORDING

- Mitochondria: "**aerobic respiration**", "provide **ATP**", "to move the flagellum". "Produce / make energy" is not accepted.
- Flagellum: "rotates / moves the sperm"; streamlined shape "**reduces resistance**".
- Longer flagellum or faster sperm: "swim faster", "**increased chance of fertilising** the egg", "competition with sperm from other males". "First to reach the egg" alone is ignored.
- Sperm head: "**haploid** nucleus, so a diploid zygote forms"; "acrosome contains **enzymes** to **digest** the zona pellucida".
- Egg: "haploid"; "**lipid** droplets as a source of energy / ATP"; "cortical granules / zona pellucida, which prevent polyspermy"; "more cytoplasm / named organelles". "Food store", "make energy" and "jelly coat" are ignored.
- Sperm drawing: acrosome, nucleus, **midpiece with mitochondria**, flagellum.

Often asked as: explain the role of the mitochondria or flagellum; draw and label a sperm; suggest why a species has longer sperm; explain how the egg is adapted; why the egg is so much larger than the sperm.

COMMON MISTAKE

"Produce energy." Use "release / provide ATP".

Saying "reach the egg" when the mark needs "**fertilise** the egg". Offering a haploid nucleus as an adaptation for **swimming**.

Placing the zona pellucida **inside** the cell membrane. Describing a feature without saying how it helps.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Sperm contain DNA in both the **nucleus** and the **mitochondria**; the acrosome's enzymes are coded by **nuclear** genes (WBI12 Jun 2019, Oct 2019).
- **Acrosin** is the enzyme in the acrosome (WBI12 Jan 2024).
- Eggs release chemicals that **attract** sperm, and have surface glycoproteins that bind them (WBI12 Oct 2021).
- Lipid droplets in large fish eggs supply energy for the developing embryo (WBI12 Jan 2023).

SPEC 3.12 Fertilisation in mammals

WHAT THIS SPEC POINT ASKS

Know the process of fertilisation in mammals: the acrosome reaction, the cortical reaction and the fusion of

nuclei.

The sequence

1. Sperm are deposited near the cervix and swim through the **uterus** into the **oviduct**, where the egg is. Chemicals released by the egg attract them.
2. A sperm passes between the follicle cells and reaches the **zona pellucida**.
3. **Acrosome reaction**: the sperm head binds to receptor glycoproteins on the zona. The **acrosome** fuses with the sperm's membrane and releases its **hydrolytic enzymes** (acrosin). The enzymes **digest a path** through the zona pellucida, and the sperm swims through.
4. The sperm's **cell membrane fuses** with the egg's cell surface membrane, and the **sperm nucleus enters** the egg.
5. **Cortical reaction**: sperm entry triggers a rise in Ca^{2+} in the egg. **Cortical granules** move to the egg membrane, **fuse** with it and release their **enzymes** (exocytosis) into the zona. The zona pellucida **hardens** and its sperm receptors are altered, so **no other sperm can bind or enter**. This prevents **polyspermy** and keeps the zygote **diploid**.
6. The egg (a secondary oocyte) now **completes meiosis II**.
7. The **haploid sperm nucleus and haploid egg nucleus fuse**, forming a **diploid zygote** (2n = 46 in humans). This fusion of nuclei is fertilisation. The zygote then divides by **mitosis** (3.17).

Don't mix up the two reactions

	Acrosome reaction	Cortical reaction
Whose?	the **sperm**	the **egg**
Trigger	sperm binding to the zona pellucida	a sperm entering the egg
Vesicles involved	the acrosome	cortical granules
Effect on the zona	**digests** a path through it	**hardens** it
Purpose	lets a sperm reach the egg membrane	stops **more** sperm entering (blocks polyspermy)

MARK-SCHEME WORDING

- Cortical reaction (3 marks): "cortical granules **fuse** with the (egg cell surface) membrane and **release enzymes**"; "zona pellucida **hardens** / thickens"; "to prevent **polyspermy** / more than one sperm entering" (so the zygote stays diploid). "Chemicals" is ignored.
- Hardening also "modifies the sperm receptors, preventing other sperm binding".

- Acrosome reaction: "acrosome **releases enzymes / acrosin**"; "enzymes **digest** the zona pellucida"; "sperm membrane **fuses** with the egg membrane (so the nucleus enters)". "Break down / penetrate" without digest is not credited.
- After entry: "**fusion of two haploid nuclei** to form a diploid nucleus".
- An egg with three nuclei: one egg nucleus + **two sperm nuclei** (polyspermy, e.g. because the zona was damaged). "Three sperm entered" is rejected.
- Low acrosin activity: the zona is digested more slowly, so fewer eggs are fertilised.

Often asked as: explain the cortical reaction; describe the events after a sperm enters; explain an egg with extra nuclei; interpret data on acrosin, ZP3 or polyspermy; evaluate an IVF technique (6-mark).

COMMON MISTAKE

Mixing up the acrosome reaction and the cortical reaction.

Leaving out the first step of the cortical reaction: the granules **fuse with the membrane and release enzymes**. "Toughened outer layer" and "fertilisation membrane" aren't precise enough.

Describing sperm entry when the question only asks what happens **after** it.

Mixing up a mammal's egg with a plant ovule (there's no triploid endosperm in mammals).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **ZP3** is the zona glycoprotein that sperm bind to. If ZP3 is missing or changed in shape, the sperm can't bind and the acrosome reaction isn't triggered (WBI12 Jun 2025).
- A **fast electrical block**: the egg's membrane potential changes within seconds of sperm entry, before the zona hardens (sea urchin data, WBI12 Jan 2026).
- **ICSI** (injecting one sperm directly into the egg) bypasses the zona, so it helps sperm with little acrosin or poor swimming; used in IVF of endangered species (WBI12 Jan 2024).
- Protein fibres (Rab27a) carry cortical granules to the membrane; without them the cortical reaction fails (WBI12 Oct 2025).

SPEC 3.13 Fertilisation in flowering plants

WHAT THIS SPEC POINT ASKS

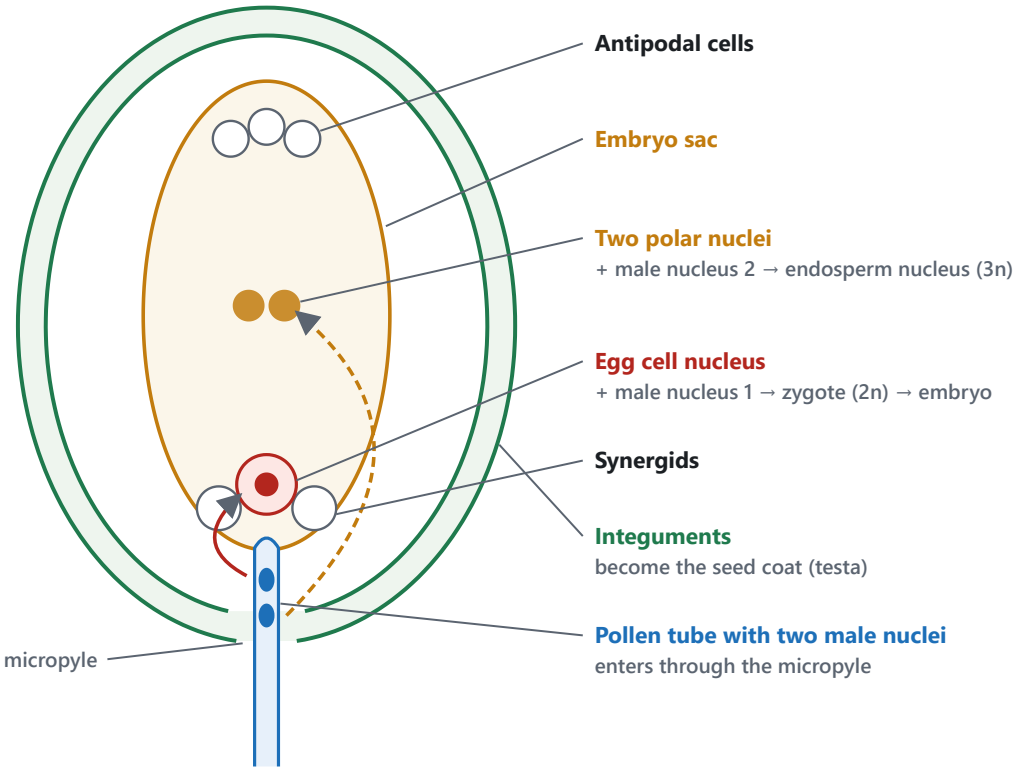
Know how fertilisation happens in flowering plants, from the growth of the pollen tube to the fusion of nuclei.

Pollination comes first, then fertilisation

- **Pollination**: transfer of pollen from an **anther** to a **stigma**. It isn't fertilisation, in the same way that mating isn't fertilisation in a mammal.
- A pollen grain is **haploid** and contains **two nuclei**: a **tube nucleus** and a **generative nucleus**. There are no sperm cells.
- The **ovule** (inside the ovary) contains the **embryo sac**: a haploid **egg cell** near the **micropyle**, **two polar nuclei** in the centre, plus synergids and antipodal cells.

The steps

1. The pollen grain lands on a compatible stigma, absorbs water and sugar, and **germinates**: a **pollen tube** grows out of it.
2. The tube grows **down through the style** towards the ovary. It secretes **hydrolytic (digestive) enzymes** that digest a path through the style tissue, and absorbs the products for growth.
3. The **tube nucleus** sits at the growing tip and **controls the tube's growth** (it holds the genes for the digestive enzymes). The tube is guided by chemicals from the ovule (chemotropism).
4. The **generative nucleus** divides by **mitosis** to form **two male nuclei** (haploid). They travel down the tube behind the tube nucleus.
5. The tube enters the ovule through the **micropyle**. The **tube nucleus breaks down**, the tip bursts, and the two male nuclei enter the embryo sac.
6. **Double fertilisation**:
    - one male nucleus **fuses with the egg nucleus** → **diploid zygote** (2n), which develops into the **embryo**;
    - the other male nucleus **fuses with the two polar nuclei** → **triploid endosperm nucleus** (3n), which divides to form the **endosperm**, the food store for the embryo.



Schematic ovule (not to scale). The pollen tube enters through the micropyle and releases two male nuclei: one fuses with the egg nucleus (zygote, 2n), the other with the two polar nuclei (endosperm, 3n).

Fusion	Product	Chromosome sets	Becomes
male nucleus + **egg nucleus**	zygote	**2n** (diploid)	embryo (the new plant)
male nucleus + **two polar nuclei**	endosperm nucleus	**3n** (triploid)	endosperm (food store in the seed)

After fertilisation the ovule becomes the **seed** (integuments → seed coat, or testa) and the ovary becomes the **fruit**.

CORE PRACTICAL

Recommended practical: factors affecting pollen tube growth

1. Make **sucrose solutions** of different concentrations (e.g. 0, 5, 10, 15, 20%), or keep one concentration and vary temperature.
2. Put a drop of each on a cavity slide and dust **pollen** from the same flower species onto it. Add a coverslip.
3. Keep the slides at a constant **temperature** (e.g. 25 °C) in a humid box.

4. After a fixed time (e.g. 30 minutes), measure the **length** of several pollen tubes with a calibrated **eyepiece graticule**, or count the percentage of grains that have germinated.
5. Repeat at intervals to find the **rate** of growth; use a **mean** of many tubes.

Controls: same species and age of flower, volume of solution, temperature, time. Sucrose supplies sugar for respiration and sets the water potential; too high a concentration draws water **out** of the grain.

MARK-SCHEME WORDING

- Pollen tube: "(tube) grows (through the style) to the ovary / ovule / embryo sac"; "(by releasing) **digestive / hydrolytic enzymes**"; "**carries** the male nuclei / generative nucleus".
- Generative nucleus: "**divides by mitosis** to form two male (haploid) nuclei". "The generative nucleus fertilises / fuses" is **rejected**.
- Double fertilisation: "one male nucleus fuses with the **egg cell nucleus** to form the zygote"; "the other male nucleus fuses with the **two polar nuclei** to form the endosperm".
- Endosperm "**3n / triploid**"; embryo cells **2n**.
- Tube nucleus: "**controls the growth** of the pollen tube"; "contains the genes for / controls production of the digestive enzymes". "The tube nucleus fertilises" is ignored.
- Data on a drug: "as the concentration increases, the length of the pollen tube decreases", plus a manipulated figure for "determine".

Often asked as: explain the role of the pollen tube (often 5 marks); the function of the tube nucleus or generative nucleus; why embryo and endosperm cells contain different amounts of DNA; identify structures on an ovule diagram; interpret pollen-tube data.

COMMON MISTAKE

Saying the **generative nucleus** fertilises the egg. It must **divide by mitosis first**; the two **male nuclei** do the fusing.

"Degenerative nucleus", "the pollen grain moves down the tube", "two sperm cells".

Writing "egg" for "egg **nucleus**", or leaving out the **polar nuclei**. Calling the endosperm haploid.

Confusing the tube nucleus with a male nucleus.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The name **double fertilisation**, the **polar nuclei**, the **micropyle** and the **embryo sac** are all expected, though the spec doesn't list them (WBI12 Oct 2019, Oct 2020, Jan 2021).
- Why the DNA differs: embryo cells have **two** sets of chromosomes (one male nucleus + egg), endosperm cells have **three** (one male nucleus + two polar nuclei) (WBI12 Jan 2024).

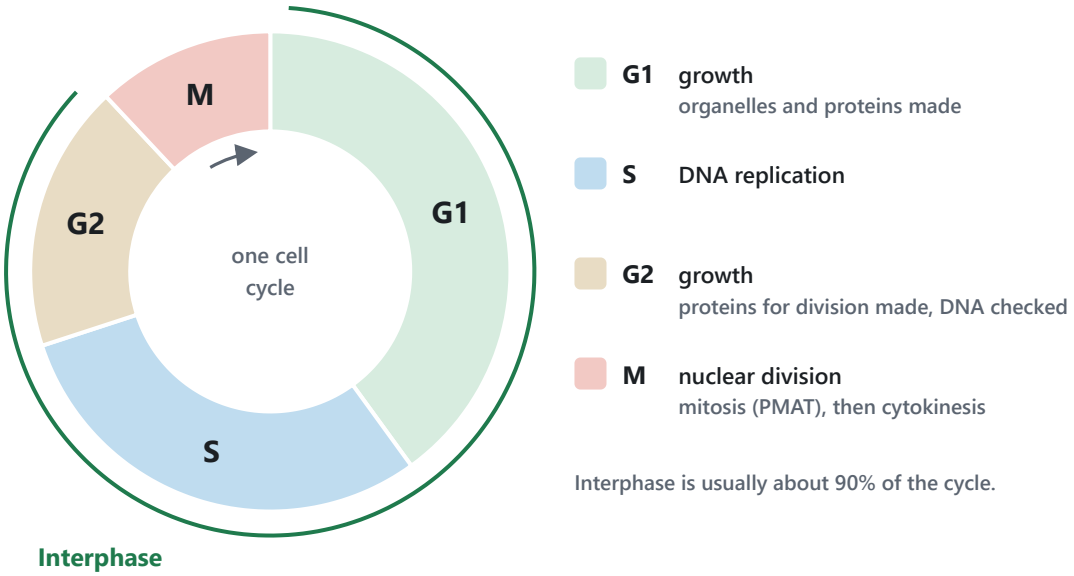
- Calcium ions are needed for pollen tube growth; a drug that blocks Ca^{2+} uptake (verapamil) shortens the tubes (WBI12 Oct 2019).
- Strategies that prevent self-fertilisation increase genetic variation but lower the chance of seeds forming (WBI12 Jun 2023, 6-mark).

SPEC 3.14 Mitosis and the cell cycle

WHAT THIS SPEC POINT ASKS

Understand how mitosis and the cell cycle produce genetically identical daughter cells for growth and asexual reproduction.

The cell cycle



The cell cycle. Interphase (G1, S, G2) takes most of the time; mitosis and cytokinesis are short. The length of each phase varies between cell types.

- **Interphase** (not part of mitosis): the cell is very active.
  - **G1**: the cell grows, makes proteins (including enzymes for DNA replication) and **makes new organelles**.
  - **S**: **DNA replicates** (semi-conservatively), so each chromosome becomes two identical **sister chromatids**.
  - **G2**: more growth, proteins for division made; the cell checks the DNA has been copied correctly.
- **Mitosis**: the nucleus divides (prophase, metaphase, anaphase, telophase).
- **Cytokinesis**: the cytoplasm divides, making two cells.

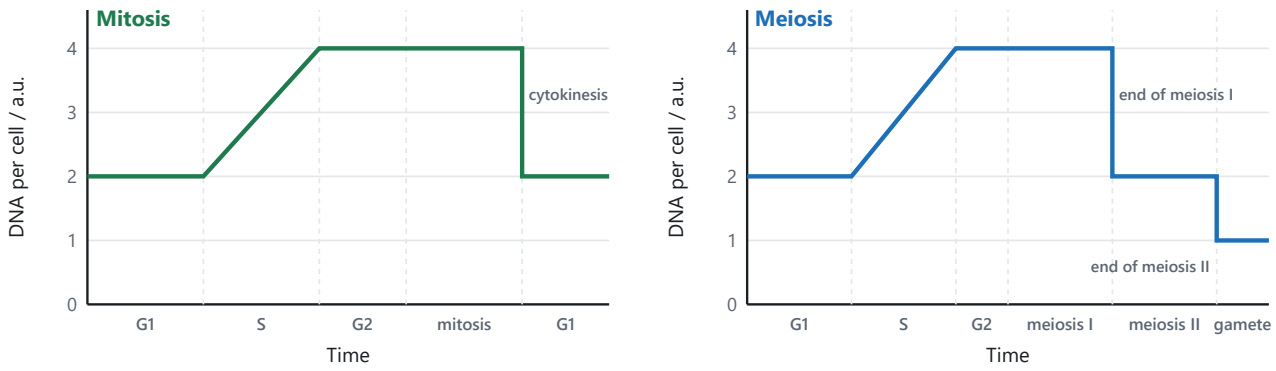
The stages of mitosis (PMAT)

Stage	What happens
Prophase	chromosomes **condense** (shorten and thicken) and become visible as two chromatids joined at a **centromere**; the **nuclear envelope** and nucleolus break down; **centrioles** move to opposite poles and the **spindle** forms
Metaphase	chromosomes line up on the **equator** of the spindle; each **centromere attaches to spindle fibres**
Anaphase	**centromeres divide**; spindle fibres **shorten** and pull the **sister chromatids** to opposite poles, centromere first
Telophase	chromatids (now chromosomes) reach the poles and **uncoil**; a **nuclear envelope** re-forms round each set; spindle breaks down
Cytokinesis	animal cells: the membrane pinches in (a **cleavage furrow**). Plant cells: a **cell plate** forms across the middle and becomes the new cell walls

Why the daughter cells are identical

1. In S phase, every chromosome is **replicated** exactly, so the two sister chromatids are identical.
2. In anaphase, the sister chromatids of **every** chromosome are separated, one to each pole.
3. So each daughter cell receives **one identical copy of every chromosome**: same number (diploid stays diploid), same alleles.

DNA content through the cycle



DNA per cell doubles during S phase, stays constant through G2 and mitosis, then halves at cytokinesis. In meiosis it halves twice: after meiosis I and after meiosis II.

What mitosis is for

- **Growth**: a zygote → embryo → adult; meristems in plants.
- **Repair and replacement**: healing wounds, replacing skin and gut lining cells, blood cells.
- **Asexual reproduction**: one parent produces genetically identical offspring (clones): budding in yeast and *Hydra*, runners in strawberries, tubers in potatoes. (Bacteria divide by **binary fission**, which isn't mitosis.)

WORKED EXAMPLE

time in a phase from a pie chart

A cell cycle lasts **24 hours**. On a pie chart, interphase is 300°, mitosis 45° and cytokinesis 15°.

1° = 24 × 60 ÷ 360 = **4 minutes**.

Mitosis: 45 × 4 = 180 min = **3 hours**. Cytokinesis: 15 × 4 = **60 minutes**.

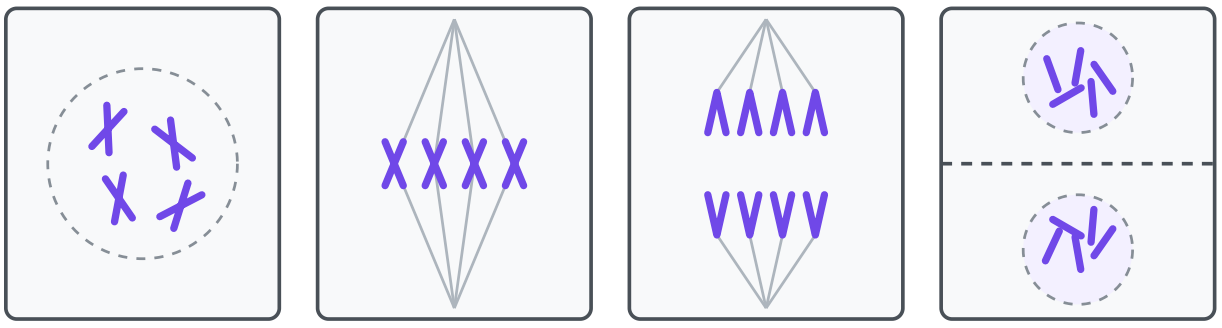
Give the unit the question asks for (hours or minutes).

WORKED EXAMPLE

cells after repeated divisions

One cell divides by mitosis 8 times, every cell dividing each time.

Cells = 2^8 = **256** (not 2 × 8 = 16).



The stages of mitosis as seen in onion root tip cells, drawn as a diagram: P (prophase), M (metaphase), A (anaphase) and T (telophase).

MARK-SCHEME WORDING

- DNA graph (4 marks): "DNA content remains **constant** during G1 / G2 / mitosis"; "DNA content **doubles** due to DNA **replication** (S phase)"; "the cell **divides**"; "producing two diploid, genetically identical daughter cells". "Interphase" alone isn't an explanation.
- Identical cells: "**derived from the zygote**" (in an embryo); "DNA / chromosomes are **replicated**"; "each cell receives one identical copy of each chromosome". "Mitosis" alone isn't enough.
- Spindle: "centromeres attach to spindle fibres"; "**separation of (sister) chromatids**"; each daughter cell gets identical genetic material.
- Drug that stops spindle fibres shortening: "(sister) chromatids **cannot be separated**" in **anaphase**.
- Role of the cell cycle: "synthesis of **organelles**", "increase in cell **size**", "**replication of DNA**". It is not a list of mitosis stages.
- Drawing anaphase for 2n = 6: spindle fibres, centrioles, **six** chromosomes at **each** pole, no nuclear envelope. Metaphase: each chromosome drawn as **two chromatids joined at a centromere**, on the equator; drawing a nuclear envelope loses the mark.
- Plant cell just after mitosis: nuclear envelopes and **cell plate** (new wall) forming.

Often asked as: order or identify stages; draw a stage for a given 2n; explain a DNA-content graph; role of the spindle or centromere; times from a pie chart; evaluate anti-cancer drug data (6-mark).

COMMON MISTAKE

Describing a graph when "explain" is asked; missing the final mark for **why DNA halves** (the cell divides).

Drawing 3 chromosomes per pole in anaphase when 2n = 6 (it's 6 at each pole). Drawing anaphase inside a nucleus.

Leaving out DNA **replication** when explaining why daughter cells are identical.

Describing the stages of mitosis when the role of the whole **cell cycle** is asked.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Cancer** is uncontrolled mitosis. Drugs that stop spindle fibres working hold cells in metaphase, so they **raise** the mitotic index of a sample while stopping division (WBI12 Oct 2022).
- **Epigenetic marks** are copied when DNA replicates, so they pass to both daughter cells (WBI12 Oct 2024; see 3.20).
- Cells of the **morula and blastocyst** divide by mitosis, not meiosis (WBI12 Jun 2024).
- Error bars on drug data: non-overlapping bars suggest a **significant** difference (WBI12 Jun 2019).

SPEC 3.15 Core Practical 6: root tip squash

WHAT THIS SPEC POINT ASKS

Prepare and stain a root tip squash and use it to observe the stages of mitosis.

Principle

- In a root, cells divide only in the **meristem**, just behind the root cap (the first 1–5 mm of the tip). That's where you'll see cells in mitosis.
- The tissue must be **separated into a layer one cell thick**, and the **chromosomes stained**, so you can see them clearly.

CORE PRACTICAL

Core practical 6: method in brief

1. Grow roots from a garlic clove or onion with its base in water for a few days.
2. Cut off the last **5–10 mm** of a few root tips.
3. Put them in warm (about 55–60 °C) **1 mol dm^{-3} hydrochloric acid** for about **5 minutes**.
4. Rinse the tips in cold water and blot them dry.
5. Put a tip on a slide, cut off and keep only the end 1–2 mm (the meristem), and add a few drops of **stain** (acetic orcein or toluidine blue). Leave a few minutes (some methods warm it).
6. Break the tip up with a mounted needle, add a coverslip, and **squash** gently by pressing down through filter paper. Don't twist the coverslip.
7. Find the meristem cells at **low power** (small, square, in rows, no large vacuoles), then view at **×400**.
8. Identify and draw cells in each stage; count cells in each stage and in interphase for the mitotic index (3.16).

Step	Why
Use the **root tip**	the meristem is where cells are dividing
Warm **HCl**	**breaks down the middle lamella** (pectin) holding cells together, so the cells separate and can be squashed flat
Rinse after the acid	removes acid, which would stop the stain working
Stain (acetic orcein / toluidine blue)	binds to DNA so the **chromosomes** are visible and the stages can be told apart (acetic orcein also fixes the cells, stopping division)
Squash	spreads the cells into a **single layer**, so light passes through and cells don't overlap

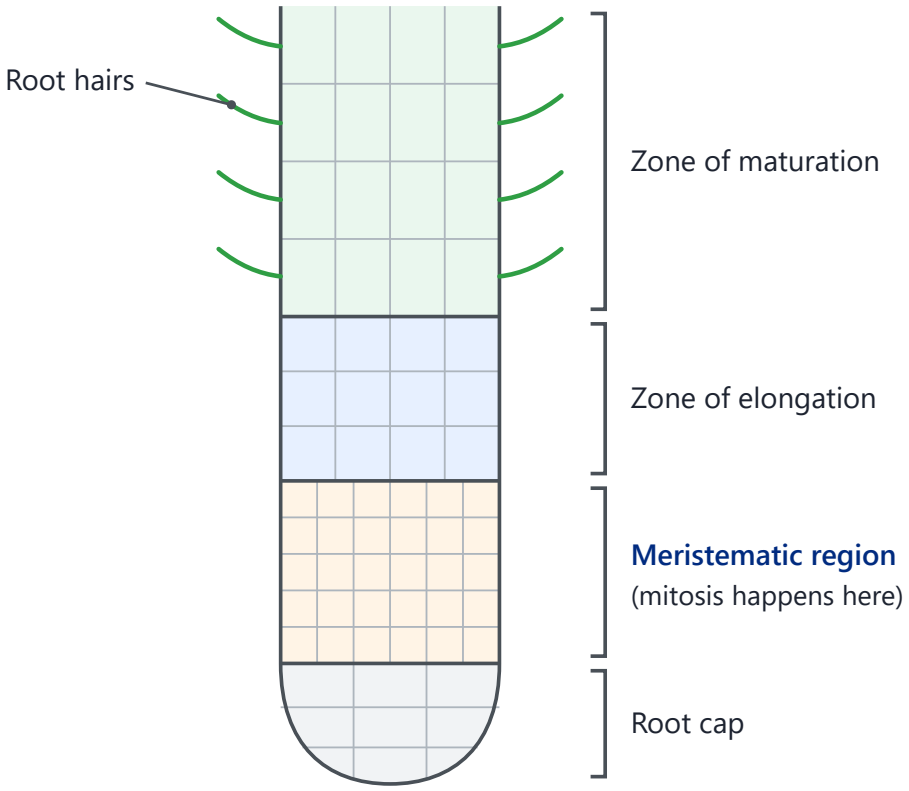
Step	Why
Low power first	to find the meristem; then high power to see chromosomes

Recognising the stages in a squash

- **Interphase**: a clear, round nucleus with a nucleolus; no visible chromosomes.
- **Prophase**: threads of chromosomes visible inside the nucleus.
- **Metaphase**: chromosomes in a line across the middle of the cell.
- **Anaphase**: two groups of V-shaped chromatids moving to opposite ends.
- **Telophase**: two dense groups at opposite ends; a new cell wall may be forming between them.

Safety and errors

- Acetic orcein is **corrosive** and stains: eye protection and gloves. HCl is an irritant at this concentration: eye protection. Hot water bath: scald risk.
- Errors: squashing too hard (chromosomes destroyed) or too lightly (cells overlap); missing the meristem; it can be hard to tell late interphase from early prophase. Count **many fields of view** and repeat with several roots.



Zones of a root tip. Mitosis happens in the meristematic region just behind the root cap.

MARK-SCHEME WORDING

- Why stain: "to make the **chromosomes / chromatids** visible", "to distinguish the stages of mitosis". "To see which cells are dividing" and "to make organelles visible" are ignored.
- Using a photo: "count the number of cells in mitosis and the total number of cells", then divide (× 100).
- Choosing a stain: the slide stained for **nucleic acids** (e.g. safranin) shows chromosomes; one stained for lipids (nile red) doesn't. Both the choice and the reason are needed.

Often asked as: the reason for the staining step (the most common question); how to find the mitotic index from a slide or photo. The full method hasn't been asked in WBI12 2019–2026, but learn it.

COMMON MISTAKE

"The stain makes the cells visible." More than half the candidates lost this mark (Jan 2020). Say **chromosomes**.

Saying the acid "softens the cell walls". It breaks down the **middle lamella** between cells, so they separate.

SPEC 3.16 Calculating mitotic indices

WHAT THIS SPEC POINT ASKS

Be able to calculate a mitotic index.

mitotic index = number of cells in mitosis ÷ total number of cells counted

- "Cells in mitosis" = cells in prophase + metaphase + anaphase + telophase (chromosomes visible).
- **Total** = cells in mitosis **+ cells in interphase**. Multiply by 100 for a percentage.
- A **high** mitotic index means cells are dividing fast: a growing region (meristem), or a **tumour**. Doctors use it to judge how aggressive a cancer is.

WORKED EXAMPLE

mitotic index

In one field of view: interphase 112, prophase 14, metaphase 4, anaphase 3, telophase 4.

Cells in mitosis = 14 + 4 + 3 + 4 = **25**. Total = 112 + 25 = **137**.

Mitotic index = 25 ÷ 137 = **0.18** (or **18.2%**).

WORKED EXAMPLE

how long is each stage?

The proportion of cells in a stage ≈ the proportion of the cycle that stage takes.

Using the counts above with a 20-hour cycle, anaphase = 3 ÷ 137 × 20 × 60 = **26 minutes**.

Anaphase is the shortest stage, so the fewest cells are caught in it.

WORKED EXAMPLE

working backwards

A tissue sample has a mitotic index of **0.24**, and 60 cells are in mitosis.

Total cells = 60 ÷ 0.24 = **250**. Cells in interphase = 250 − 60 = **190**.

Getting a reliable value

- Count at least 100 cells, in **several fields of view**, from several root tips; calculate a mean.
- Use a clear rule for cells cut by the edge of the field (e.g. count those on two edges only).
- Count cells at the **same distance from the tip** when comparing treatments.

Calculating the mitotic index

1. Count the total number of cells in one field of view.

   Use the same field and magnification for every count so the results can be compared.

2. Count the cells that show visible chromosomes, which means cells in prophase, metaphase, anaphase or telophase.

3. Divide the number in mitosis by the total, then multiply by 100 to get a percentage.

   A percentage makes counts from different fields directly comparable.

MARK-SCHEME WORDING

- "Number of cells in mitosis ÷ total number of cells (× 100)". 0.22 and 22% are both accepted.
- Method: "count the number of cells in mitosis (visible chromosomes)"; "divide by the **total** number of cells"; "× 100".

- A treatment that stops chromosomes condensing: "fewer cells in mitosis", "**lower** mitotic index", "higher survival".
- Higher mitotic index → more advanced cancer stage and lower survival.

Often asked as: calculate an index from counts (show working); rearrange to find the total or interphase cells; describe how to find the index from a slide; predict survival from an index.

COMMON MISTAKE

Dividing by the number of **interphase** cells instead of the **total**. This is the most common error every year.

Adding up the total wrongly; getting a number of cells larger than the total; forgetting × 100 when a percentage is asked.

Quick checks

QUICK CHECK

In which stage of mitosis do sister chromatids separate and move to opposite poles?

- Prophase
- Metaphase
- Anaphase
- Telophase

Anaphase is when the centromeres divide and the chromatids are pulled apart. Metaphase is when chromosomes line up, and telophase is when they arrive at the poles and decondense, so both are common wrong answers.

QUICK CHECK

A human cell has 46 chromosomes before meiosis. How many chromosomes does each gamete contain?

- 46
- 92
- 23
- 12

Meiosis halves the chromosome number, so each gamete has 23 single chromosomes. The answer 46 is the common mistake of forgetting that meiosis reduces the number. The answer 92 is the number of chromatids present after DNA replication, before division, so it is a trap for students who mix up chromosomes and chromatids.

QUICK CHECK

Which process produces genetic variation during meiosis I?

- DNA replication in interphase
- Anaphase of mitosis
- Crossing over between homologous chromosomes in prophase I
- Cytokinesis in plant cells

Crossing over exchanges sections between homologous chromatids in prophase I, creating new allele combinations. DNA replication copies DNA without changing it, and the other two options produce identical cells, so they are the wrong answers.

QUICK CHECK

A root tip has 200 cells in view, and 30 of them show visible chromosomes. What is the mitotic index?

- 3.0
- 15
- 15.0 %
- 30

Divide 30 by 200 and multiply by 100, which gives 15.0 %. Leaving out the multiplication by 100 gives 0.15, which is the common error.